Mass Balance Equations for Retention and Basis Weight in a Paper Machine

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***Abstract*— In a paper industry, especially in thick stock wet end process, the two most important variables are retention in forming section of a paper machine and basis weight of the thick stock. Present paper attempts to understand the governing mass balance equations which may be required to deduce an appropriate process model.**

Keywords: Process model; Paper machine; Mass balance equations; Modeling

I. INTRODUCTION

In practice, the most common variables of interest to measure are the headbox flow rate, the consistencies of various white water streams flowing through the forming fabric, and the consistency of the paper on the top of the forming fabric. Flow of stock as well as white water can be measured in the sections of headbox and wire. Regardless of which definition is used to calculate first-pass total solids retention or ash retention, it is absolutely critical that the consistencies and ash levels must be determined with good precision. On-line real time SISO feedback control of recycled white water consistency can be a way to retention control. Thick stock wet end chemistry processes and machine direction basis weight variation play key roles in this regard. Hence papermaker's control strategy should be to minimize the variation in the recycled white water consistency [1].

Small errors in the headbox and tray consistency and ash data can produce large variation in the calculated first pass retention values. [2]. More advanced closed loop retention control system using two consistency sensors, the headbox and white water tray and a computational module has also been proposed [3].

The sensors were capable of determining both the total consistency and ash consistency of the sampled streams. It has been found that the total recycled white water consistency measurement provided the quickest response to a process disturbance and would therefore, be the best control variable. To circumvent large variations that affect retention should be dealt with in the thick stock area, before the furnish reaches to the basis weight valve [4, 5].

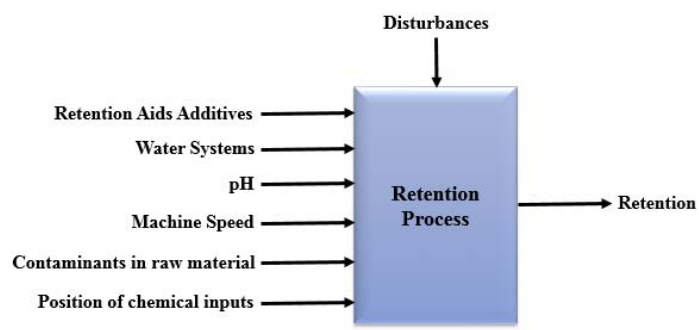

Fig. 1. Variables affecting retention process

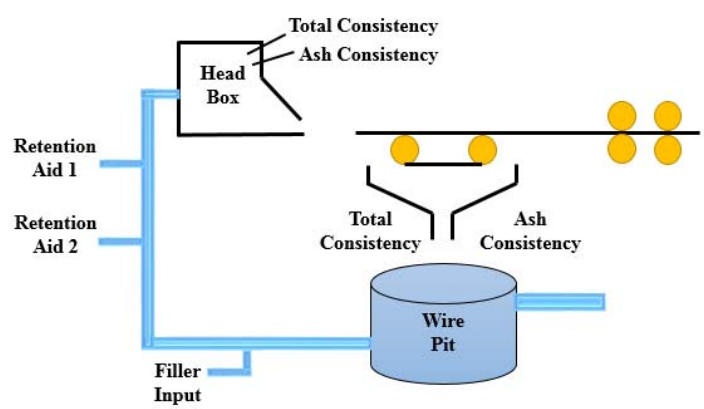

Fig. 2. Main parts of retention control in the wet end

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The retention of a substance is the amount of substance that remains in the sheet of paper, expressed as a fraction of that in the paper at some earlier stage in the process. The first pass retention of a material represents the fraction of material that passes through the headbox slice and is retained during the sheet formation. First pass retention is of extreme importance to the papermaker because it influences paper machine runnability, material loss levels, and chemical use efficiencies.

It is very important to note that retention data is in each case derived from a basic calculation used in paper machine. Retention on a paper machine is calculated on the basis of the head box and white water consistencies. It cannot be measured directly by any system, but it is calculated between two or more measuring positions.

First-pass retention values are usually derived from the steady-state material balance conditions existing around a forming section. Since first-pass retention is a steady-state parameter, it is not strictly correct to refer to it when a paper machine is not operating under steady-state conditions. In such instances, it is often more useful simply to monitor the key parameters involved, like headbox consistency and tray consistency. The variables affecting retention process are shown in Fig. 1 and Fig.2 demonstrates the key parts of wet end and retention control.

II. Governing Equations for Retention

Solids flow on paper machine can be represented by Fig. 3.

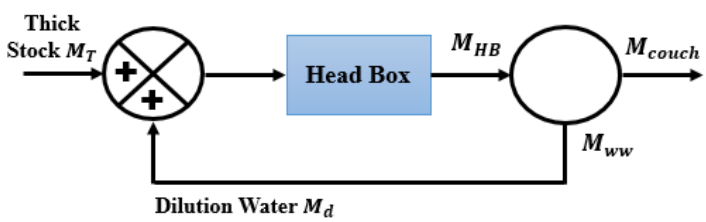

Fig. 3. Solids flow on paper machine

$$M_{HB} = M_T + M_d \tag{1}$$

$$Q_{HB}C_{HB}\rho = Q_T C_T \rho + Q_d C_d \rho \tag{2}$$

Or

$$Q_{HB}C_{HB} = Q_T C_T + Q_d C_d$$

Using volume balance

$$Q_d = Q_{HB} - Q_T$$

We get

$$C_{HB} = \frac{1}{Q_{HB}}[Q_T C_T + (Q_{hb} - Q_T)C_d] \tag{3}$$

The equation for CHB give the value of thin stock consistency produced by given thick stock volume flow rate and consistency, thin stock flow rate and dilution water consistency. Thin stock consistency is inversely proportional to thin stock flow rate provided the through factor does not change.

$$M_{HB} = M_T + M_d$$

As

$$M_d = M_{ww} - M_{wl}$$

$$M_{HB} = M_T + M_{ww} - M_{wl} \tag{4}$$

Also

$$M_{ww} = TM_{HB} \tag{5}$$

$$M_{wl} = LM_{HB} \tag{6}$$

$$M_{HB} = M_T + TM_{HB} - LM_{HB} \tag{7}$$

$$M_{HB}(1 - T + L) = M_T$$

$$M_{HB} = \frac{M_T}{(1 - T + L)} \tag{8}$$

Putting

$$M_T = Q_T C_T \rho \text{ and } M_{HB} = Q_{HB} C_{HB} \rho$$

We get

$$Q_{HB}C_{HB} = \frac{Q_T C_T}{(1 - T + L)} \tag{9}$$

$$C_{HB} = \frac{Q_T C_T}{Q_{HB}(1 - T + L)}$$

Or

$$Q_{HB} = \frac{Q_T C_T}{C_{HB}(1 - T + L)} \tag{10}$$

If L is negligible in comparison to (1-T), then

$$Q_{HB} = \frac{I}{C_{HB}}\left(\frac{Q_T C_T}{1 - T}\right) \tag{11}$$

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And

$$Q_{HB} = \frac{I}{C_{HB}}\left(\frac{Q_T C_T}{1-T}\right) \quad (12)$$

III. Governing Equations for Basis Weight

$$Total\ Solids = M_{couch} = M_{HB} - M_{ww} \quad (13)$$

$$M_{couch} = M_{HB} - TM_{HB} \quad (14)$$

$$M_{couch} = M_{HB}(1-T) \quad (15)$$

Also,

$$M_{couch} = WGV_w \quad (16)$$

$$WGV_w = M_{HB}(1-T) \quad (17)$$

$$G = \frac{M_{HB}(1-T)}{WV_w}$$

As

$$1 - T = R$$

$$G = \frac{M_{HB}R}{WV_w} \quad (18)$$

Also

$$M_{couch} = M_T - M_{bw}$$

$$M_{couch} = M_T - LM_{HB} \quad (19)$$

Where

$$M_{bw} = M_T - \frac{LM_T}{(1-T+L)}$$

As

$$M_{HB} = \frac{M_T}{(1-T+L)}$$

$$M_{couch} = \frac{M_T(1-T)}{(1-T+L)} = \frac{M_T}{\left\{1+\frac{L}{(1-T)}\right\}} \quad (20)$$

$$WGV_w = \frac{M_T}{\left\{1+\frac{L}{(1-T)}\right\}}$$

$$G = \frac{M_T}{WV_w\left\{1+\frac{L}{(1-T)}\right\}} \quad (21)$$

$$G = \frac{M_T}{WV_w} \quad if\ L \ll (1-T) \quad (22)$$

As

$$M_T = Q_T C_T \rho$$

$$G = \frac{Q_T C_T \rho}{WV_w\left\{1+\frac{L}{(1-T)}\right\}} \quad (23)$$

Or

$$Q_T = \frac{GWV_w}{C_T \rho}\left\{1+\frac{L}{(1-T)}\right\}$$

This gives the thick stock volume flow rate required to produce the desired value of grammage at given value of wire speed.

IV. Conclusion

Mass balance equations corresponding to basis weight and retention for a paper machine are presented which can lead to a suitable process model.

Nomenclature

V_{HB}	=	head-box tank volume
Q_T	=	volume flow rate of thick stock (m^3/s)
ρ_T	=	thick stock density
V	=	total stock volume
P	=	total stock density
I_f	=	Ignition factor
Q_t	=	flow rate of thin stock
Q_{ww}	=	mass flow rate of white water
Q_T	=	thick stock volume flow rate
Q_r	=	stock retained mass flow rate
Q_{in}	=	stock entering wire volume flow rate
A_T	=	total stock ash amount
A_{ww}	=	white water ash amount
A_t	=	thick stock ash amount
Q_{TLw}	=	white water drained solids volume flow rate
Q_{Tr}	=	retained solids volume flow rate
W	=	slice width
G	=	basis weight
V_w	=	speed of wire

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T	=	through factor
W_m	=	width of machine
V_s	=	velocity of slice jet
Q_{bw}	=	flow rate of backwater
C_{bw}	=	consistency of backwater
K	=	constant of recirculation
E_R	=	ratio of efflux
h	=	slice pressure head
C	=	discharge coefficient
A_H	=	ash level in headbox
A_{ww}	=	ash level in white water
f_{HB}	=	fines content in head box
f_{ww}	=	fines content in white water
DS_{HB}	=	dissolved solids content in headbox
DS_{ww}	=	dissolved solids content in white water

Flow Rates:

M_s	=	thick stock mass
M_d	=	dilution rate
M_o	=	thick stock
M	=	solid
M_{Tin}	=	forming section Total solid
M_{Tww}	=	white water drained solid
M_{Tr}	=	retained solids
M_{Ta}	=	accumulated solids
M_{couch}	=	couch solids
M_{wl}	=	white water losses mass
M_{bb}	=	breast box

Volumetric flow rate:

q_s	=	thick stock
q_o	=	thin stock
q_d	=	dilution water

Consistency:

C_s	=	Consistency
C_T	=	total stock
C_{ww}	=	white water
c_s	=	thick stock
c_d	=	dilution water
c_o	=	thin stock
C_{mp}	=	main pump outlet stock
C_{bp}	=	balance pipe stock
C_{couch}	=	in fraction for couch
C_{HB}	=	in fraction for headbox
C_{ww}	=	in fraction for white water

Retention:

γ_a	=	of ash
γ_f	=	of fines
γ_{ds}	=	of dissolved solids

REFERENCES

[1] W. E. Scott, "Principles of Wet End Chemistry," *Tappi Press*, 1995.

[2] K. Leiviska, "Paper making science and technology- Process control Book," *Finnish Paper Engineers, Association and TAPPI,*1999.

[3] J. Nolelainen, R. Piirainen, "Practical Experiences of white water consistency control of a paper machine wet end," *In Papermakers Conference, Tappi press,* pp. 227-238, 1993.

[4] O. J. Antonio, J. Paris, M. Perrier, "Paper machine controllability: effect of disturbances on basis weight and first-pass retention," *Journal of process control*, vol. 11, no. 4, pp. 401-408, 2001.

[5] P. K. Juneja, M. Chaturvedi, S. Suman, K. Antil, "Modeling of stock consistency in the approach flow system of the headbox", *In 2018 3rd International Conference On Internet of Things: Smart Innovation and Usages (IoT-SIU), IEEE*, Bhimtal, India, 2018, pp. 1-4.

[6] P. K. Juneja, M. Chaturvedi, A.K. Ray, G. Yadav, "Modeling for headbox and associated wet end systems," *In Proceedings of ICETIT 2019,* Springer, Cham pp. 396-406, 2020.

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